



POLITECNICO

MILANO 1863

Financial Engineering

Assignment 3

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Individual Solution

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0. Introduction

This report presents the solutions to five exercises spanning core risk-management and derivatives-pricing techniques: Value-at-Risk and Expected Shortfall estimation for linear equity portfolios via parametric (Variance-Covariance), Historical Simulation, Weighted Historical Simulation, Bootstrap, and PCA approaches (Exercises 0–1); Full Monte-Carlo and Delta-Normal VaR for a portfolio with a non-linear payoff (Exercise 2); pricing of a Cliquet option under counterparty risk via CVA adjustment (Exercise 3); and pricing of CDO tranches under the Vasicek model, comparing exact, Large Homogeneous Portfolio (LHP), and Kullback-Leibler approximations (Exercise 4). All methods are validated against the theoretical framework presented in the course.

1. Exercise 0 — Variance–Covariance VaR & ES

Setup

We consider an equally weighted portfolio (notional $V = \text{€}10,000,000$) invested in Inditex, BASF and LVMH. Daily log-returns are estimated over a 2-year window (26 July 2010 – 23 July 2012, $T = 516$ observations), with missing quotes filled using the previous available price, since the three stocks do not trade on exactly the same calendar days. We estimate the 1-day, 95% VaR and Expected Shortfall as of 24 July 2012 under the Variance–Covariance (parametric Gaussian) approach.

Methodology

Under the Gaussian assumption, portfolio log-returns are $r_{ptf} \sim \mathcal{N}(\mu_{ptf}, \sigma_{ptf}^2)$, with $\mu_{ptf} = w^\top \mu$ and $\sigma_{ptf}^2 = w^\top \Sigma w$, where μ and Σ are the sample daily mean vector and covariance matrix of the historical returns. VaR and ES at confidence level α then follow in closed form:

$$VaR_\alpha = V(-\mu_{ptf} + z_\alpha \sigma_{ptf}), \quad ES_\alpha = V\left(-\mu_{ptf} + \sigma_{ptf} \frac{\phi(z_\alpha)}{1 - \alpha}\right), \quad (1)$$

where $z_\alpha = \Phi^{-1}(\alpha)$ and $\phi(\cdot)$ is the standard Normal density.

Estimated inputs

Table 1: Daily mean log-returns and volatilities (equal weights $w_i = 1/3$).

Asset	Weight	Daily mean return μ_i	Daily volatility $\sqrt{\Sigma_{ii}}$
Inditex	1/3	7.932×10^{-4}	1.577%
BASF	1/3	3.598×10^{-4}	1.926%
LVMH	1/3	4.501×10^{-4}	1.851%

Table 2: Sample daily correlation matrix.

	Inditex	BASF	LVMH
Inditex	1.000	0.489	0.541
BASF	0.489	1.000	0.712
LVMH	0.541	0.712	1.000

The resulting portfolio daily volatility is $\sigma_{ptf} = 1.521\%$.

Results

Comments

As expected, $ES_{95\%} > VaR_{95\%}$, consistent with the Expected Shortfall being the average loss beyond the VaR quantile. The VaR represents approximately 2.45% of the portfolio notional, in line with the

Table 3: Exercise 0: 1-day, 95% VaR and ES (Variance–Covariance method).

Risk measure	Value (€)
VaR _{95%}	244,867.63
ES _{95%}	308,431.42

estimated portfolio daily volatility (1.52%) scaled by the 95% Normal quantile ($z_{0.95} \approx 1.645$). LVMH and BASF show the highest pairwise correlation (0.71), which limits the diversification benefit relative to Inditex, whose correlation with the other two names is comparatively lower (≈ 0.49 – 0.54).

2. Exercise 1 — Historical, Weighted Historical, Bootstrap and PCA VaR

As of 24 July 2012, an asset manager computes risk measures for three portfolios at the $\alpha = 99\%$ confidence level, using a 2-year (516-observation) historical window of daily log-returns.

2.1. Portfolio 1 — Historical Simulation & Bootstrap

Portfolio 1 holds 18,000 ENI, 25,000 Telefónica, 15,000 EON and 9,000 Daimler shares, valued at the reference date, giving a notional of $V = \text{€}1,058,870.00$ and weights $w = (0.2592, 0.2038, 0.2299, 0.3071)$. We compute the 1-day, 99% VaR and ES via (i) Historical Simulation (HS), taking the largest integer not exceeding $n(1 - \alpha)$ as the VaR order statistic, and (ii) a Statistical Bootstrap with $M = 200$ resamples with replacement (random seed fixed at 42 for reproducibility), applied jointly to all risk factors to preserve their dependence structure.

Table 4: Portfolio 1: 1-day, 99% VaR and ES.

Method	VaR (€)	ES (€)
Historical Simulation	54,612.87	61,064.05
Statistical Bootstrap ($M = 200$)	60,054.75	66,641.37

Table 5: Portfolio 1: Plausibility checks.

Method	VaR (€)
Correlation-aggregation Plausibility Check	46,739.54
Univariate Gaussian Plausibility Check	40,190.54

Both HS and Bootstrap VaR/ES are of the same order of magnitude as the two plausibility checks, confirming there are no gross errors in the implementation; the Bootstrap estimate is higher than HS, an effect of the additional resampling noise from only $M = 200$ draws. A Jarque-Bera test on the portfolio's historical returns (skewness -0.23 , kurtosis 4.73 , $p < 0.001$) rejects normality, which justifies preferring the non-parametric HS/Bootstrap approaches over a purely Gaussian one for this portfolio.

2.2. Portfolio 2 — Weighted Historical Simulation

Portfolio 2 is an equally weighted portfolio ($w_i = 1/5$) of Vivendi, AXA, Enel, Volkswagen and Schneider Electric. Since the assignment does not specify a notional for this portfolio, we assume $V = \text{€}10,000,000$. We compute the 1-day, 99% VaR and ES via Weighted Historical Simulation (WHS), with exponential decay factor $\lambda = 0.98$, which assigns higher weight to more recent observations. The VaR order i^* is taken as the largest scenario index whose cumulative weight (ordered from worst to best loss) does not exceed the tail budget $1 - \alpha$, consistent with the definition given in class.

The WHS VaR and ES are lower than the correlation-aggregation plausibility check but of the same order of magnitude, and closer to the univariate Gaussian check; this is consistent with WHS placing

Table 6: Portfolio 2: 1-day, 99% VaR and ES (WHS, $\lambda = 0.98$).

Method	VaR (€)	ES (€)
Weighted Historical Simulation	401,826.63	481,403.79

Table 7: Portfolio 2: Plausibility checks.

Method	VaR (€)
Correlation-aggregation Plausibility Check	506,680.15
Univariate Gaussian Plausibility Check	445,707.58

most of the weight on the most recent observations, while the plausibility checks use unweighted historical percentiles. As for Portfolio 1, a Jarque-Bera test (skewness -0.002 , kurtosis 5.12 , $p < 0.001$) rejects normality of the portfolio's return distribution.

2.3. Portfolio 3 — Gaussian Parametric PCA

Portfolio 3 is an equally weighted portfolio of the first 25 EUROSTOXX 50 constituents (up to and including France Télécom), with notional $V = \text{€}15,000,000$. We compute the 10-day, 99% VaR via a Gaussian parametric PCA approach with n principal components, and compare it against the full Gaussian (Variance–Covariance) VaR to determine the minimum n achieving a relative error below 5% and below 1%.

Table 8: Portfolio 3: Full Gaussian VaR and PCA order thresholds.

Quantity	Value
Full Gaussian VaR (10-day, 99%)	€1,845,103.55
Min. n for relative error $\leq 5\%$	1
Min. n for relative error $\leq 1\%$	1

Table 9: PCA-approximated VaR and relative error vs. number of components n .

n	PCA VaR (€)	Relative error
1	1,857,491.75	0.671%
2	1,848,434.78	0.181%
3	1,848,441.62	0.181%
4	1,849,833.95	0.256%
5	1,849,089.63	0.216%
6	1,846,997.49	0.103%
7	1,848,256.12	0.171%
8	1,846,497.90	0.076%

Table 10: Portfolio 3: Plausibility checks.

Method	VaR (€)
Correlation-aggregation Plausibility Check	2,040,664.05
Univariate Gaussian Plausibility Check	1,787,477.73

Remarkably, a *single* principal component already reduces the relative error below both the 5% and 1% thresholds (0.67%). This indicates that the return covariance structure of these 25 large-cap Eurozone stocks is dominated by one common risk factor (a broad equity-market factor), so that most of the portfolio's variance is already captured by the first eigenvalue/eigenvector pair; adding further

components only marginally refines the estimate. As for the other two portfolios, a Jarque-Bera test (skewness -0.08 , kurtosis 4.88 , $p < 0.001$) rejects normality of the realized portfolio returns, even though the risk measure itself is computed under a Gaussian assumption; this should be mentioned as a caveat on the PCA method's accuracy: precisely *because* normality is rejected, the Gaussian-based VaR (both the full Variance–Covariance estimate and its PCA approximation) is expected to *underestimate* the true tail risk of Portfolio 3, since the Normal distribution has thinner tails than the empirical one.

3. Exercise 2 — Full Monte-Carlo and Delta-Normal VaR for a Non-Linear Portfolio

Setup

At the reference date of 15 February 2010 (adjusted to the nearest business day), the portfolio consists of n shares of Generali (notional €1,164,000) and the same number n of European put options on Generali stock, with strike $K = €28.50$, maturity 18 April 2010 (adjusted to 19 April 2010, a Monday), annualized volatility $\sigma = 22.3\%$, and continuous dividend yield $q = 5.1\%$. Given the observed stock price $S_0 = €16.55$ on the reference date, the position size is $n = \text{round}(1,164,000/S_0) = 70,332$. The risk-free rate $r = 4.247\%$ is obtained by interpolating the bootstrapped zero-coupon curve at the option's time to maturity ($T = 63$ days $= 0.1726$ years). We compute the 1-day, 99% VaR via a Full Monte-Carlo (full-valuation historical simulation) approach and a Delta-Normal (linear) approximation, both driven by a 2-year historical window of Generali daily log-returns (15 February 2008 – 15 February 2010, 512 observations).

Methodology

Under Full Monte-Carlo, each of the $T = 512$ historical daily returns is applied as a shock to today's stock price, $S_{sim} = S_0 e^{r_{hist}}$, the portfolio is fully repriced (reducing time to maturity by one day) via Black–Scholes, and VaR is extracted as the order statistic $[T(1 - \alpha)]$ of the resulting loss distribution. Under the Delta-Normal approach, only the first-order (Delta) sensitivity of the portfolio is used:

$$VaR_\alpha = -\Delta_\$ \mu_h + |\Delta_\$| \sigma_h z_\alpha, \quad (2)$$

where $\Delta_\$ = (n + n \delta_P) S_0$ is the portfolio's dollar delta w.r.t. the underlying's log-return, δ_P is the put's Black–Scholes delta, and μ_h, σ_h are the historical daily mean and volatility of Generali's log-returns scaled to the 1-day horizon.

Results

Table 11: Position and pricing details.

Quantity	Value
Stock price S_0	€16.55
Put price P_0	€11.89
Put delta δ_P	-0.9912
Put gamma γ_P	≈ 0.00 (deep in-the-money)
Shares / puts held (n)	70,332
Portfolio value V_0	€2,000,022.79
Portfolio delta (shares)	616.40

Comments

The put is deep in-the-money ($S_0 = 16.55 \ll K = 28.5$), so its delta is close to -1 and its gamma is essentially zero. As a result, one share plus one put form an almost delta-neutral combination (portfolio

Table 12: Exercise 2: 1-day, 99% VaR.

Method	VaR (€)
Full Monte-Carlo (full valuation)	518.67
Delta-Normal approximation	489.47

delta of 616 shares out of 70,332, i.e. under 1% net exposure), which explains why the absolute VaR is small relative to the €2 M notional ($\approx 0.026\%$). Because gamma is negligible here, the gap between the two methods (€518.67 vs. €489.47, about 6%) is *not* primarily driven by unhedged convexity, but by the fact that Full Monte-Carlo uses the actual empirical distribution of Generali's returns over 2008–2010 — a period that includes the global financial crisis and is markedly fat-tailed and left-skewed — while the Delta-Normal method assumes a Gaussian return distribution. Full valuation therefore captures a heavier realized left tail than the Gaussian approximation predicts, so the Delta-Normal method *understates* risk here.

Delta vs. Delta-Gamma Approximation (ATM case)

To visualize how well the linear (Delta) and quadratic (Delta-Gamma) approximations track the true Black–Scholes put price, and how their error grows away from the current spot, Figures 1 and 2 report both on an at-the-money case ($S_0 = K = \text{€}28.5$, same maturity, volatility and rate as above) rather than on the actual deep in-the-money position, since gamma — and hence the benefit of a second-order correction — is largest near the money.

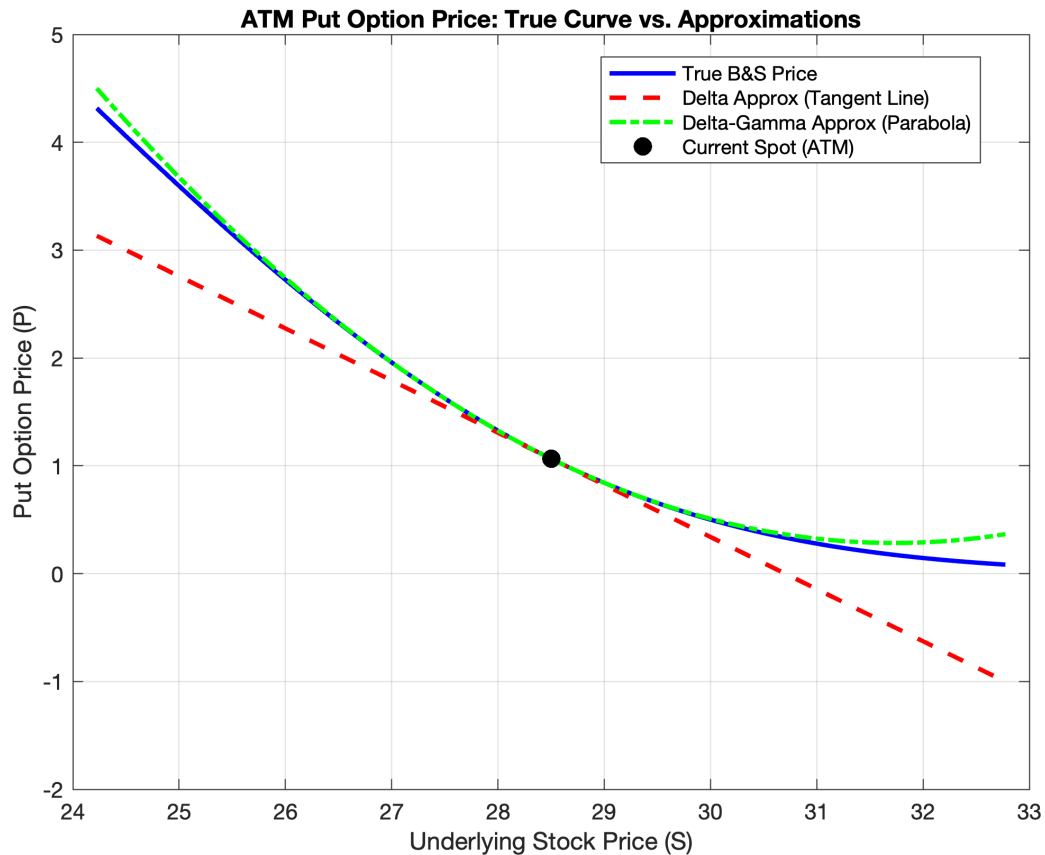


Figure 1: ATM put price: true Black–Scholes curve vs. its Delta (tangent line) and Delta-Gamma (parabola) approximations around $S_0 = K = \text{€}28.5$.

As expected, the Delta (linear) approximation diverges quickly and asymmetrically from the true price as $|\Delta S|$ grows, while the Delta-Gamma (quadratic) approximation tracks the true curve closely over

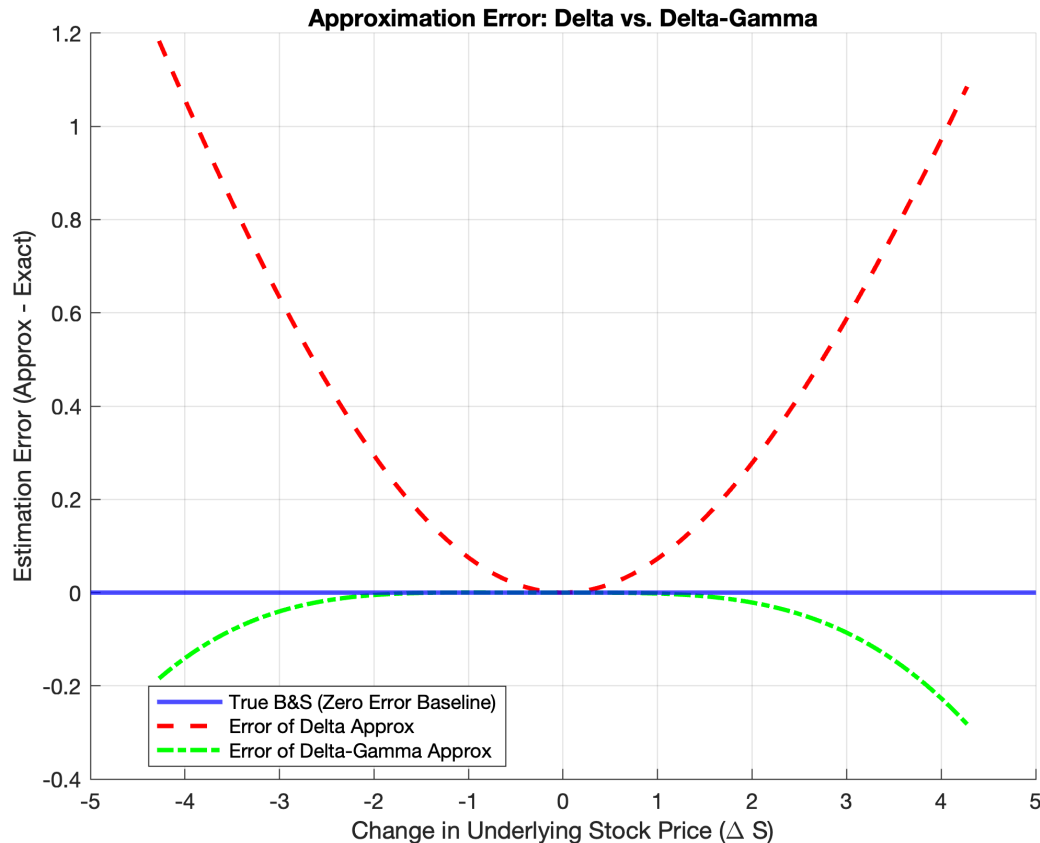


Figure 2: Approximation error (Approx – Exact) of the Delta and Delta-Gamma loss approximations as a function of the change in the underlying stock price, ATM case.

a much wider range, confirming that gamma captures the bulk of the curvature of the put's payoff near the money. This is precisely the effect that is *absent* in the actual Exercise 2 position: since that portfolio is deep in-the-money ($S_0 = 16.55 \ll K = 28.5$), gamma is essentially zero (see the first table above) and the Delta approximation is already adequate from a convexity standpoint — the gap observed between Full Monte-Carlo and Delta-Normal VaR is therefore attributable to the non-Gaussian (fat-tailed) empirical return distribution rather than to unhedged convexity, as discussed above.

Can the Delta-Normal VaR be improved? How?

Since gamma is negligible for this particular position, extending the approximation to second order (Delta-Gamma) would bring little benefit here. The relevant improvement is instead on the *distributional* side: replacing the Gaussian assumption on the risk factor with its empirical historical distribution, while keeping the linear (Delta) approximation for repricing — i.e. a *Delta-Historical* VaR. This retains the computational simplicity of a linear approximation (no repricing needed) while capturing the fat left tail that the Full Monte-Carlo result shows to be material. More generally, for positions with non-negligible gamma, a Delta-Gamma (Cornish-Fisher or Monte-Carlo) approach would also correct for convexity.

Why can Full Monte-Carlo be numerically intensive for an exotic derivative without a closed-form price?

Full Monte-Carlo requires repricing the *entire* portfolio under every one of the T scenarios (and, for a genuine forward-looking Monte-Carlo, under every one of many simulated paths). For a vanilla option this repricing is a single, closed-form Black-Scholes evaluation per scenario — essentially free. For an exotic derivative with no closed-form price, each scenario's valuation instead requires its own

numerical procedure (e.g. a nested Monte-Carlo simulation, a PDE solver, or a tree), so the total cost scales as (number of risk scenarios) $\times$ (cost of one exotic pricing) — potentially a simulation nested inside a simulation. This can become prohibitively expensive as the number of scenarios and/or the resolution of the inner pricing procedure grow.

4. Exercise 3 — Cliquet Option Pricing under Counterparty Risk

Setup

On 15 February 2008 at 10:45 CET, bank XX buys from ISP a 5-year Cliquet option on ISP stock (no dividends, $q = 0$, constant volatility $\sigma = 19\%$) with notional $M = \text{€}45,000,000$. The option pays annually, at each t_i ($i = 1, \dots, 5$), the reset (ATM) payoff $[S(t_i) - S(t_{i-1})]^+$, i.e. a percentage-return caplet on the previous year's realized stock performance. As is standard for this data set, the reference date used for discounting is the curve's settlement date, 19 February 2008. Since ISP (the option seller) may default before honoring its obligations, bank XX (the buyer) is exposed to counterparty risk and must price the option net of a Credit Value Adjustment (CVA).

For the CVA, we bootstrap ISP's survival probability curve from CDS spreads of 29, 34, 37, 39 and 40 bps for the 1–5 year tenors, with a constant recovery rate $R = 40\%$, using the discrete-premium (accrual-neglected) bootstrap convention. These market inputs are reused, for illustrative purposes, from a related assignment specifically focused on CDS bootstrapping (which also compared this approximate convention against the exact accrual-adjusted and Jarrow–Turnbull methods); they are not part of the present assignment's data set.

Methodology

Cliquet pricing.

Since each caplet resets at-the-money at the start of its accrual period, its time-0 value follows from the tower property of conditional expectation:

$$V_i = S_0 e^{-qT_{i-1}} \cdot C_{BS}\left(1, 1, r_i^{fwd}, q, \sigma, \delta_i\right), \quad (3)$$

where T_{i-1} is the time from the valuation date to the start of period i , $\delta_i = T_i - T_{i-1}$ is the accrual length, and r_i^{fwd} is the continuously compounded forward rate for $[T_{i-1}, T_i]$ bootstrapped from the discount curve. This correctly prices the option using the true risk-neutral dynamics of the underlying stock (rather than of the corresponding forward contract), as required by the assignment. The total theoretical price is $TP = M \sum_{i=1}^5 V_i$.

CVA.

Since all Cliquet cash flows are non-negative and (by assumption) independent of the default time τ , the CVA of the full cash flow stream decomposes additively, caplet by caplet, via the tower property:

$$CVA = (1 - R) M \sum_{i=1}^5 V_i \cdot \mathbb{Q}(\tau \leq t_i), \quad (4)$$

where $\mathbb{Q}(\tau \leq t_i) = 1 - Q_{surv}(t_i)$ is ISP's cumulative default probability up to the i -th payment date, bootstrapped from the CDS curve above. The risk-adjusted price is then Correct Price = $TP - CVA$.

Results

Comments

Each caplet is worth roughly 9.3%–9.7% of notional, consistent with a series of 1-year at-the-money calls at 19% volatility and a $\approx 4\%$ risk-free rate; the theoretical price of the full Cliquet, at $\text{€}21.45$ M, is about 47.7% of the $\text{€}45$ M notional. The CVA represents only 1.13% of the theoretical price, consistent with a relatively low and stable ISP CDS curve (29–40 bps) over the trade's 5-year life and a 40% recovery assumption.

Table 13: Cliquet caplet values (per unit notional) and CDS-implied survival inputs.

i	Payment date	Forward rate r_i^{fwd}	Caplet value V_i
1	19-Feb-2009	3.931%	0.095235
2	19-Feb-2010	3.649%	0.093609
3	21-Feb-2011	3.860%	0.095014
4	20-Feb-2012	4.089%	0.095747
5	19-Feb-2013	4.309%	0.097070

Table 14: Cliquet price and CVA adjustment.

Quantity	Value
Theoretical price (TP)	€21,450,358.40
CVA	€241,402.44
Risk-adjusted (correct) price	€21,208,955.96
CVA as % of TP	1.13%

What should be the correct price?

Since bank XX is the option buyer and therefore bears the risk that ISP (the seller) defaults before the option's cash flows are paid, the price bank XX should be willing to pay is the counterparty-risk-adjusted price, Correct Price = $TP - CVA = €21,208,955.96$.

At what price would ISP try to sell it?

ISP, as the seller, is not exposed to its own default risk in the same way (from its own pricing perspective it would not deduct a CVA against itself); it would therefore try to sell the option at the full theoretical price $TP = €21,450,358.40$, i.e. without the counterparty risk discount that only matters to the buyer. The €241,402.44 gap between the two figures is exactly the price of ISP's own credit risk, and is the natural object of negotiation between the two parties.

5. Exercise 4 — CDO Tranche Pricing under the Vasicek Model

Setup

The reference portfolio of SPV Cayman II has a total notional of €1 bn, assumed homogeneous: each obligor defaults independently of interest rates within $T = 3$ years with individual default probability $PD = 5\%$, asset correlation $\rho = 40\%$, and recovery rate $R = 20\%$. We price the mezzanine tranche ($K_d = 5\%$, $K_u = 9\%$) and, as a facultative extension, the equity tranche ($K_d = 0\%$, $K_u = 5\%$), both as a percentage of tranche face value, under the single-factor Gaussian copula (Vasicek/ASRF) model, for a number of obligors I ranging from 10 to 2×10^4 on a logarithmic grid.

Methodology

Conditional on the systematic factor $y \sim \mathcal{N}(0, 1)$, each obligor's default probability is

$$p(y) = \Phi\left(\frac{\Phi^{-1}(PD) - \sqrt{\rho}y}{\sqrt{1-\rho}}\right), \quad (5)$$

and, conditional on y , the number of defaults M follows a $\text{Binomial}(I, p(y))$ distribution. Writing the fractional loss as $z = M/I$, the tranche loss (as a percentage of tranche face value) is

$$L(z) = \min\left(\max\left(\frac{(1-R)z - K_d}{K_u - K_d}, 0\right), 1\right), \quad (6)$$

and the tranche price is $P = DF(0, T) \cdot (1 - \mathbb{E}[L(z)])$, where the expectation is computed under three alternative models:

- **Exact:** $\mathbb{E}[L(z)] = \int_{-\infty}^{\infty} \phi(y) \sum_{m=0}^I \binom{I}{m} p(y)^m (1-p(y))^{I-m} L(m/I) dy$, computed directly from the Binomial pmf. This is only numerically feasible up to the point where $\binom{I}{m}$ loses floating-point precision (in our implementation, around $I \approx 58$).
- **LHP** (Large Homogeneous Portfolio, $I \rightarrow \infty$): by the law of large numbers, $z \rightarrow p(y)$ almost surely, and a change of variable gives the closed-form Vasicek density

$$f(z) = \sqrt{\frac{1-\rho}{\rho}} \frac{\phi(\bar{y}(z))}{\phi(\Phi^{-1}(z))}, \quad \bar{y}(z) = \frac{\Phi^{-1}(PD) - \sqrt{1-\rho} \Phi^{-1}(z)}{\sqrt{\rho}}, \quad (7)$$

so that $\mathbb{E}[L(z)] = \int_0^1 f(z) L(z) dz$. This limit does not depend on I .

- **KL (Kullback–Leibler) approximation:** for finite but large I , the Binomial pmf is replaced by its large-deviations (saddlepoint) approximation,

$$f(z | y) \approx \sqrt{\frac{I}{2\pi z(1-z)}} e^{-I \cdot KL(z||p(y))}, \quad KL(z||p) = z \ln \frac{z}{p} + (1-z) \ln \frac{1-z}{1-p}, \quad (8)$$

integrated numerically over y to obtain the marginal density $f(z)$. For the equity tranche ($K_d = 0$), the integral is started at $z = 1/(2I)$ rather than 0, since the continuous KL density is singular exactly at the origin while the true discrete distribution has a genuine probability mass at “zero defaults”.

Results

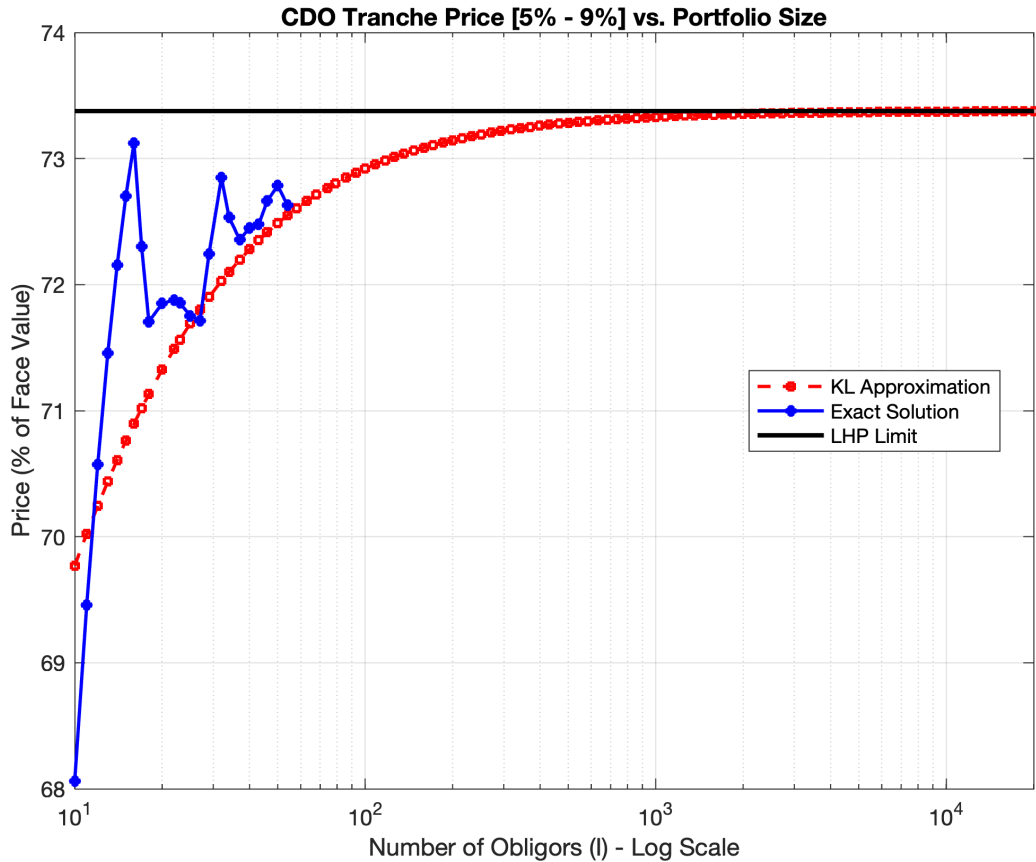


Figure 3: Mezzanine tranche [5%–9%] price vs. number of obligors I : Exact, KL approximation and LHP limit.

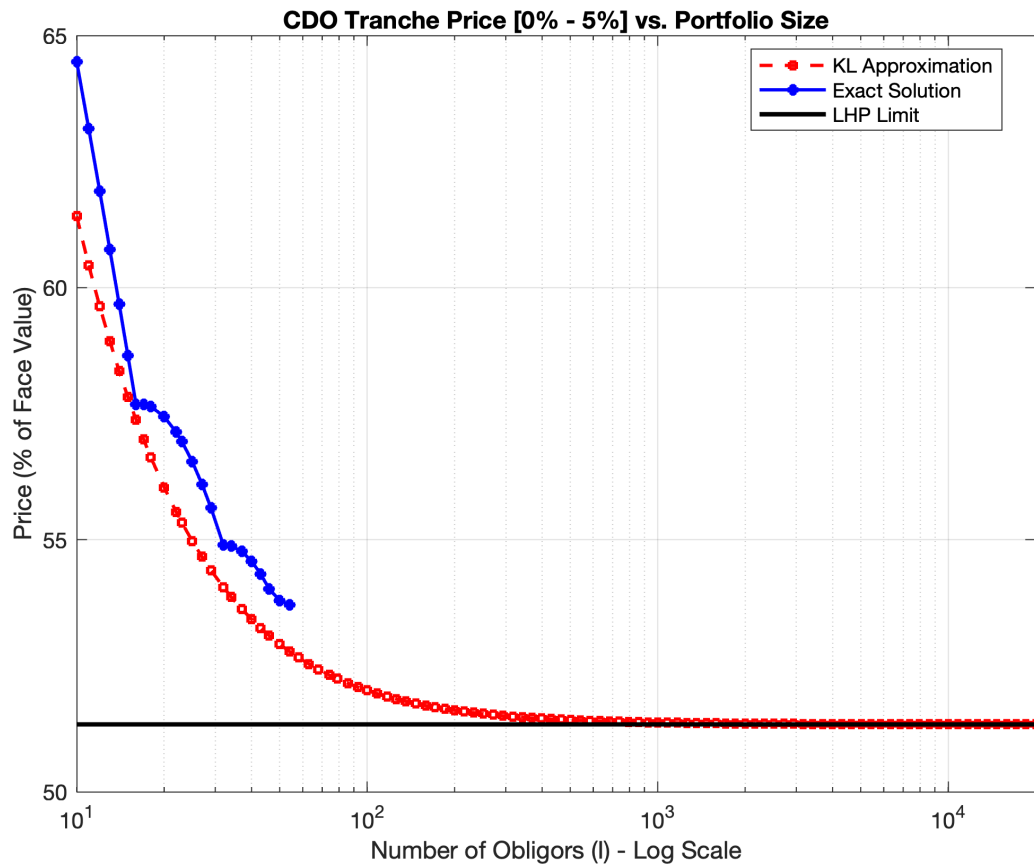


Figure 4: Equity tranche [0%–5%] price vs. number of obligors I : Exact, KL approximation and LHP limit.

Table 15: LHP limit prices (% of tranche face value).

Tranche	LHP price
Mezzanine [5%–9%]	73.377%
Equity [0%–5%]	51.337%

Comments

Both figures confirm the theoretical prediction: the exact solution (computable only up to $I \approx 58$ before $\binom{I}{m}$ loses floating-point precision) and the KL approximation converge monotonically to the LHP limit as $I \rightarrow \infty$, validating the LHP approximation exactly on the range $I \in (10, 2 \times 10^4)$ specified in the assignment. For the mezzanine tranche, the KL approximation is already within 0.16% of the LHP limit at $I = 400$ and within 0.06% at $I = 1,000$.

For the equity tranche, the KL approximation converges to the same limit but *more slowly*: at $I = 50$ its relative error is 3.1% (vs. 1.2% for the mezzanine), and it only drops below 1% around $I \approx 100$ –200. This is expected, since the equity tranche is a first-loss piece and is therefore far more sensitive to idiosyncratic (small- I) granularity effects than a mezzanine tranche further from the attachment point.

Is the KL approximation adequate for the Equity tranche? How could it be modified?

The KL approximation is qualitatively adequate for the equity tranche — it converges to the correct LHP limit and captures the right shape of the price-vs- I curve — but it is quantitatively less accurate than for the mezzanine tranche at small-to-moderate I , precisely in the region where the equity tranche is most exposed to name-specific risk. This can be improved by: (i) using higher-order

Table 16: KL approximation vs. LHP limit: relative error by portfolio size I .

I	KL Mezz. price	Rel. error	KL Equity price	Rel. error
50	72.487%	1.213%	52.927%	3.097%
100	72.920%	0.623%	52.010%	1.311%
400	73.260%	0.159%	51.453%	0.226%
1,000	73.330%	0.064%	51.374%	0.072%
5,000	73.368%	0.013%	51.343%	0.011%
20,000	73.375%	0.003%	51.339%	0.002%

saddlepoint/Edgeworth corrections to the leading-order KL (large-deviations) approximation, which are known to improve accuracy near the boundary of the loss distribution; (ii) applying a finer continuity correction than the single half-step ($z = 1/(2I)$) used here, e.g. explicitly matching the discrete probability mass at $z = 0$ (zero defaults) rather than approximating it via the continuous density; or (iii), for the range of I where it remains numerically feasible ($I \lesssim 58$ in our implementation), simply using the exact Binomial computation instead of the approximation.